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MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

Linux– Theory Assignment 6

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Cybersecurity Log Analysis Using Linux Tools

1. Introduction

Cybersecurity log analysis is the process of examining system log files to identify suspicious activities and possible security threats.

In this assignment, SSH authentication logs are analysed to detect possible brute-force login attempts.

A brute-force attack occurs when an attacker repeatedly tries different passwords to gain unauthorized access to a system.

2. Objective

The main objectives of this assignment are:

- To understand the structure of cybersecurity log files.
- To identify repeated failed login attempts.
- To detect suspicious IP addresses.
- To use Linux commands such as grep, sort, uniq, sed and awk.
- To develop a Bash script to automate log analysis

3. Log File Acquisition

A simulated SSH authentication log file was created in the Kali Linux environment.

The file was named auth.log and contained 13 sample log records.

The records included:

- Successful SSH login attempts.
- Failed SSH login attempts.
- Attempts involving invalid usernames.
- Repeated login attempts from different IP addresses.

The simulated records were created manually using Linux terminal commands. No real systems were accessed or tested.

```
┌──(sanjanarao㉿sanjanarao)-[~/Cybersecurity_Log_Analysis]
└─$ cat auth.log
Nov 05 09:10:12 kali sshd[1201]: Accepted password for user1 from 192.168.1.1
0 port 50120 ssh2
Nov 05 09:12:15 kali sshd[1202]: Failed password for invalid user admin from
203.0.113.10 port 50121 ssh2
Nov 05 09:12:20 kali sshd[1203]: Failed password for invalid user admin from
203.0.113.10 port 50122 ssh2
Nov 05 09:12:25 kali sshd[1204]: Failed password for invalid user admin from
203.0.113.10 port 50123 ssh2
Nov 05 09:12:30 kali sshd[1205]: Failed password for root from 203.0.113.10 p
ort 50124 ssh2
Nov 05 09:13:10 kali sshd[1206]: Accepted password for user2 from 192.168.1.1
5 port 50125 ssh2
Nov 05 09:15:10 kali sshd[1207]: Failed password for test from 198.51.100.20
port 50126 ssh2
Nov 05 09:15:15 kali sshd[1208]: Failed password for test from 198.51.100.20
port 50127 ssh2
Nov 05 09:15:20 kali sshd[1209]: Failed password for test from 198.51.100.20
port 50128 ssh2
Nov 05 09:15:25 kali sshd[1210]: Failed password for root from 198.51.100.20
port 50129 ssh2
Nov 05 09:15:30 kali sshd[1211]: Failed password for root from 198.51.100.20
port 50130 ssh2
Nov 05 09:16:10 kali sshd[1212]: Accepted password for user3 from 192.168.1.2
0 port 50131 ssh2
Nov 05 09:20:10 kali sshd[1213]: Failed password for guest from 192.168.1.50
port 50132 ssh2
```

```
┌──(sanjanarao㉿sanjanarao)-[~/Cybersecurity_Log_Analysis]
└─$ wc -l auth.log
13 auth.log
```

4. Log File Structure

Each log record contains the following fields:

Field	Description
Date and time	When the event occurred
Hostname	Name of the computer
sshd	SSH service generating the log
Process ID	Identifier of the SSH process
Event	Accepted or Failed password
Username	Account used in the attempt
IP address	Source address of the connection
Port	Source connection port
ssh2	SSH protocol version indicator

5. Cybersecurity Investigation Questions

1. How many SSH login attempts failed?
2. Which IP addresses generated repeated failed login attempts?
3. Were there any successful login attempts in the log file?

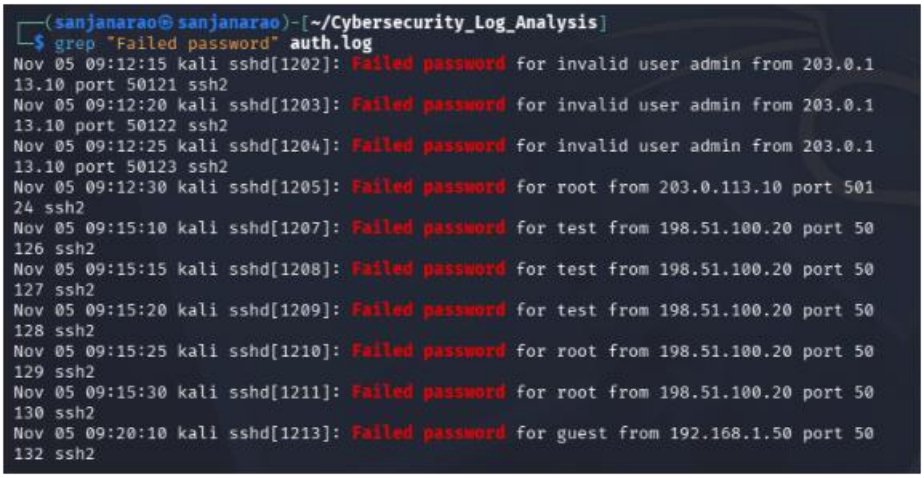
These questions help identify possible brute-force attacks and understand login activity.

6. Practical Log Analysis

Filtering Failed Login Attempts

The grep command is used to search for specific patterns in a log file. In this step, it filters all records containing "Failed password" to identify unsuccessful SSH login attempts.

Result: The command displays all failed SSH login records.



Filtering Successful Login Attempts

This command filters successful SSH login attempts from the log file. It searches for the phrase "Accepted password" to identify users who successfully logged in.

Result: The command displays all successful SSH login records.

```
┌──(sanjanarao㉿sanjanarao)-[~/Cybersecurity_Log_Analysis]
└─$ grep "Accepted password" auth.log
Nov 05 09:10:12 kali sshd[1201]: Accepted password for user1 from 192.168.1.10 port
50120 ssh2
Nov 05 09:13:10 kali sshd[1206]: Accepted password for user2 from 192.168.1.15 port
50125 ssh2
Nov 05 09:16:10 kali sshd[1212]: Accepted password for user3 from 192.168.1.20 port
50131 ssh2
```

Counting Failed Login Attempts

The -c option counts the number of lines matching a particular pattern. This command calculates the total number of failed SSH login attempts in the log file.
Result: The command displays the total number of failed login records, which is 10.

```
┌──(sanjanarao㉿sanjanarao)-[~/Cybersecurity_Log_Analysis]
└─$ grep -c "Failed password" auth.log
10
```

Extracting IP Addresses Using AWK

The grep command first filters failed login records. The awk command then extracts the source IP address from each record.
The pipe (|) operator passes the output of grep to awk. NF-3 represents the fourth field from the end of each log record.
Result: The command displays the IP addresses associated with failed login attempts.

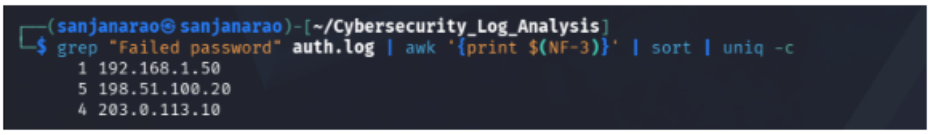
```
┌──(sanjanarao㉿sanjanarao)-[~/Cybersecurity_Log_Analysis]
└─$ grep "Failed password" auth.log | awk '{print $(NF-3)}'
203.0.113.10
203.0.113.10
203.0.113.10
203.0.113.10
198.51.100.20
198.51.100.20
198.51.100.20
198.51.100.20
198.51.100.20
192.168.1.50
```

Counting Failed Attempts from Each IP Address

This command identifies IP addresses responsible for failed login attempts and counts how frequently each address appears.

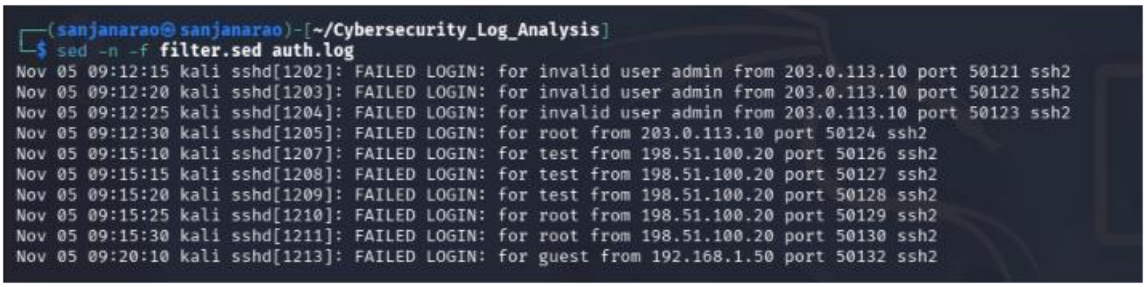
- grep filters failed login records.
- awk extracts the IP addresses.
- sort arranges the IP addresses.
- uniq -c counts repeated IP addresses.

Result: The analysis identifies IP addresses with repeated failed login attempts. This helps detect potentially suspicious login activity.



Filtering Log Records Using sed

The sed command is a stream editor used to search and modify text in a file.
In this step, a separate sed script named filter.sed was created to identify failed login records and replace the phrase "Failed password" with "FAILED LOGIN:".
The -n option suppresses automatic printing, while -f executes the commands stored in the sed script.
Result: The command displays the filtered login records with a modified message, making failed login activities easier to identify.



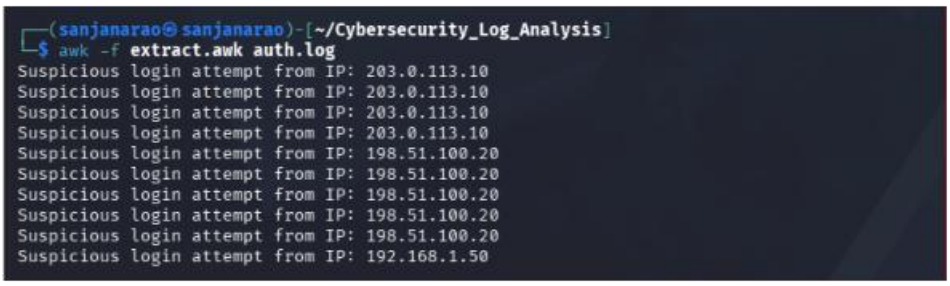
Extracting IP Addresses Using an AWK Script

AWK is a text-processing tool used to extract and analyse specific fields from log records.

A separate script named extract.awk was created to identify failed login records and extract their source IP addresses.

The script uses the pattern "Failed password" to select matching records and $(NF-3) to extract the IP address.

Result: The script displays the IP addresses associated with failed login attempts.



7. Bash Script Development

A Bash script named analyze.sh was developed to automate the SSH log analysis process.

The script combines different Linux commands, including grep, wc, sed, awk, sort and uniq.

The script performs the following operations:

1. Checks whether the authentication log file exists.
2. Counts the total number of log records.
3. Identifies failed and successful login attempts.
4. Uses sed to filter and modify failed login records.
5. Uses awk to extract source IP addresses.
6. Counts failed login attempts from each IP address.
7. Identifies suspicious IP addresses based on a threshold of three or more failed attempts.

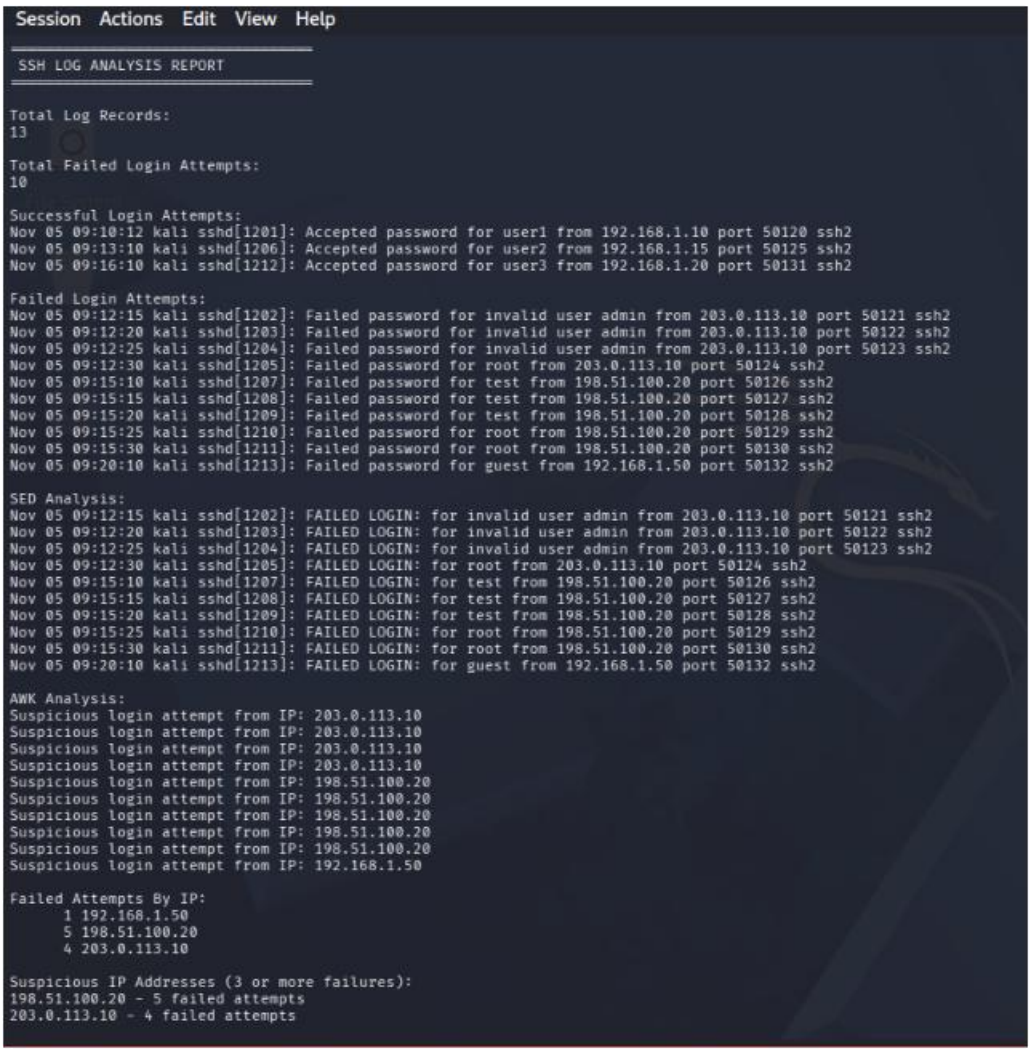
8. Saves the generated analysis report into output.txt.

The script reduces manual effort by performing multiple log analysis operations automatically.

Script Execution

The script was made executable using the chmod command and executed using the Linux terminal.

The generated results were displayed in the terminal and saved in a separate output file named output.txt.



8. Analysis Results

The Bash script was executed successfully to analyse the simulated SSH authentication log file.

The analysis produced the following results:

- Total log records: 13
- Failed login attempts: 10
- Successful login attempts: 3

The source IP addresses responsible for failed login attempts were identified and counted.

Source IP Address	Failed Attempts
198.51.100.20	5
203.0.113.10	4
192.168.1.50	1

The results show that two IP addresses generated repeated failed login attempts.

The generated analysis was also saved in a separate file named output.txt.

9. Identification of Suspicious Activities

The analysis identified two IP addresses with repeated failed SSH login attempts.

The IP address 198.51.100.20 generated five failed login attempts, while 203.0.113.10 generated four failed login attempts.

A threshold of three or more failed login attempts from the same IP address was selected to classify repeated login activity as suspicious.

Both IP addresses exceeded this threshold and were flagged for further investigation.

The IP address 192.168.1.50 generated only one failed attempt and was not classified as suspicious.

These results indicate possible brute-force activity, but they do not confirm an actual attack.

10. False Positives and Limitations

False Positives

A false positive occurs when normal user activity is incorrectly identified as suspicious.

Examples include:

- A genuine user repeatedly entering an incorrect password.
- A user forgetting their login credentials.
- Multiple users sharing the same public IP address.
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Limitations

- The analysis uses simulated log data instead of real authentication logs.
- The threshold of three attempts may not identify every attack.
- Repeated failed attempts do not always indicate malicious activity.
- The analysis only examines the available log records.
- The script does not automatically block suspicious IP addresses.

11. Conclusion

This assignment demonstrated the use of Linux command-line tools for cybersecurity log analysis.

A simulated SSH authentication log was analysed using grep, pipelines, sed, awk and Bash scripting.

The analysis successfully identified failed login attempts and extracted IP addresses responsible for repeated failures.

The Bash script automated the analysis and generated a separate output report.

This practical exercise helped demonstrate how Linux tools can be used to identify potentially suspicious activities in system logs.